



Ai-Driven Fraud Detection in Financial Transactions – Using Machine Learning and Deep Learning to Detect Anomalies and Fraudulent Activities in Banking and E-Commerce Transactions

Md Mainul Islam^{1*}, Ismoth Zerine², Md Arifur Rahman³, Md Saiful Islam⁴,
Md Yousuf Ahmed⁵

^{1*,2,3,4,5}College of graduate and professional studies, Trine university, Angola, Indiana, USA.

Email: ²izerine23@my.trine.edu, ³mrahman22@my.trine.edu, ⁴mislam21@my.trine.edu,
⁵Yousufraj92@gmail.com

Corresponding Email: ^{1*}mislam23@my.trine.edu

ARTICLE INFO

Received: 20-10-2024
Accepted: 29-11-2024

ABSTRACT

Financial fraud has become a growing concern with the rapid digitalization of banking and e-commerce platforms. Detecting fraud effectively is challenging due to the high imbalance between legitimate and fraudulent transactions. This research addressed the issue by analyzing publicly available datasets using AI-based techniques to identify patterns and indicators of fraudulent activity. The significance of the study lies in its potential to support financial institutions in minimizing fraud losses and improving transaction security without developing new models. The main objective was to evaluate and compare the performance of machine learning (ML) and deep learning (DL) methods in detecting fraudulent transactions using real-world data. Three datasets were used: the European Credit Card Fraud Dataset, PaySim (a mobile money simulator), and a dataset from the UCI Machine Learning Repository. The methodology included data cleaning, balancing using SMOTE, feature engineering, and statistical analysis. Algorithms such as Logistic Regression, Random Forest, XGBoost, and Neural Networks were applied. Model performance was assessed using accuracy, precision, recall, F1-score, and AUC-ROC. The results showed that XGBoost achieved the highest performance with an accuracy of 99.2%, precision of 96.8%, recall of 94.5%, F1-score of 95.6%, and AUC-ROC of 0.987. Deep learning models also performed well, especially on large datasets, but required more computational power. In conclusion, ML models, particularly XGBoost, were highly effective in detecting fraudulent transactions. The main finding of this research is that AI-based techniques can significantly enhance fraud detection

without complex model development. This study supports the practical implementation of existing AI methods to improve financial security and reduce fraud-related losses.

Keywords: Anomaly Detection, Deep Learning, Fraud Detection, Machine Learning, XGBoost.

INTRODUCTION

Background

In recent years, the surge in online transactions, particularly in banking and e-commerce, has been accompanied by a rise in fraudulent activities. Fraud detection in financial systems, therefore, has become one of the most critical concerns for financial institutions and e-commerce platforms (Mutemi & Bacao, 2024). With the growing sophistication of fraudsters, traditional methods of fraud detection are no longer sufficient to keep pace with the complexities of modern financial transactions (Bello et al., 2023). These systems primarily relied on rule-based approaches, which were effective in some scenarios but failed to adapt to the evolving nature of fraudulent schemes. As a result, fraud detection systems have become increasingly reliant on machine learning (ML) and deep learning (DL) techniques to identify suspicious activities and anomalies within vast amounts of transaction data (Gandhar et al., 2024).

Machine learning models are particularly effective in detecting fraud due to their ability to learn from historical data, identify patterns, and make predictions on unseen data. Deep learning, a subset of machine learning, has further revolutionized this field by providing a means to learn complex, non-linear relationships within large datasets (Dargan et al., 2020). These advancements have enhanced fraud detection capabilities, making them more adaptive and capable of detecting subtle, previously undetectable fraud patterns. As financial institutions and e-commerce platforms increasingly move toward digital transactions, it is crucial to implement more robust and scalable fraud detection systems to safeguard financial transactions (Udeh et al., 2024).

Importance of this Research

The importance of fraud detection in financial transactions cannot be overstated. Fraudulent transactions not only result in significant financial losses for individuals and businesses but also undermine the trust and integrity of the entire financial system (Udeh et al., 2024; Obeng et al., 2024). In 2022, global financial fraud losses were estimated at over \$40 billion, with the figures steadily increasing each year (Ogbebor, 2023). Fraud detection systems that are not sophisticated enough to identify emerging fraud schemes expose financial institutions to even greater risks, particularly in the context of e-commerce, where transactions are often anonymous and harder to scrutinize (Khurana, 2020).

Given the prevalence of fraud in the digital space, effective fraud detection systems are necessary for maintaining consumer confidence and protecting both businesses and consumers from financial harm. The implementation of artificial intelligence (AI), particularly machine learning and deep learning, can help address this problem by automating and enhancing the process of fraud detection, ensuring that financial transactions remain secure and trustworthy (Nizioł, 2021). This research aims to explore the potential of AI-driven fraud detection systems that can analyze vast amounts of transaction data to detect

anomalies and flag fraudulent activities in real-time, reducing human intervention and improving the overall accuracy and efficiency of fraud detection (Bello et al., 2023).

Why We Did This Research

The motivation for this research stemmed from the increasing sophistication of fraudulent activities and the limitations of existing fraud detection methods. Traditional systems, which relied on rule-based algorithms and manual input, struggled to keep up with the rapid changes in fraudulent tactics (Njoku et al., 2024). As fraudsters continuously evolve their methods to bypass security measures, the need for more dynamic and intelligent detection systems has become urgent.

Moreover, with the rapid growth in the volume and complexity of financial transactions, manual review of all transactions for fraud has become impractical. Financial institutions and e-commerce platforms require automated systems that can analyze massive datasets quickly and accurately to identify suspicious activities (Li et al., 2021). Existing machine learning models have shown promise in this area, but their deployment has often been limited by a lack of understanding regarding the most effective techniques for fraud detection. By leveraging deep learning methods, this research aimed to explore advanced solutions that could handle more complex patterns of fraud and offer enhanced performance over traditional methods (Ejiofor, 2023).

This research not only aimed to improve the detection capabilities of fraud detection systems but also sought to enhance the efficiency of the process, making it more scalable to handle the increasing volume of financial transactions (Adeleke et al., 2022). Additionally, the research aimed to address the gaps in current literature by testing and evaluating machine learning and deep learning methods within the context of both banking and e-commerce transactions, which have not been adequately compared in previous studies.

Significance of the Study

This study holds significant value for both academic research and industry applications. For academic researchers, it provides a comprehensive exploration of various machine learning and deep learning techniques applied to the problem of fraud detection in financial transactions (Alghofaili et al., 2020). By testing multiple algorithms and frameworks, the study contributes to the existing body of knowledge regarding the effectiveness of these techniques in real-world scenarios, offering valuable insights into their strengths and limitations. The results of this study can inform future research on fraud detection systems and help researchers develop more effective models (Blonde et al., 2018).

From an industry perspective, this study provides valuable insights into the application of AI-driven fraud detection methods in real-world banking and e-commerce platforms (Emran & Rubel, 2024). With the increasing prevalence of fraud, the development and implementation of more effective and accurate fraud detection systems are critical (Huang et al., 2017). The findings from this research can help financial institutions and e-commerce platforms implement more robust security measures to safeguard their transactions, protect customers, and minimize financial losses (Ahmad et al., 2023). Furthermore, as the financial sector moves toward digital transformation, the need for scalable, automated solutions becomes ever more pressing. This study's findings can guide the development of such systems, ensuring they are both efficient and effective in identifying fraudulent activities.

Research Gap

Despite the advancements in fraud detection technologies, there is still a considerable gap in the literature when it comes to comparing and evaluating the performance of machine learning and deep learning models in detecting fraud across various financial sectors. While many studies have explored the use of machine learning for fraud detection, there remains a lack of consensus on the most effective algorithms, the challenges of applying these techniques at scale, and their adaptability to different types of fraud (Ali et al., 2022). Furthermore, the majority of existing research tends to focus on either banking transactions or e-commerce, without addressing the cross-sector challenges in fraud detection.

Additionally, much of the existing research has focused on either supervised or unsupervised learning techniques, without adequately exploring hybrid approaches that combine the strengths of both methods (Azevedo et al., 2024). This research aims to bridge these gaps by exploring the performance of different machine learning models (such as decision trees, random forests, and support vector machines) alongside deep learning techniques (like convolutional neural networks and recurrent neural networks) in detecting fraud across both banking and e-commerce datasets.

Objective

The primary objective of this research was to explore and compare the effectiveness of machine learning and deep learning algorithms in detecting fraud in financial transactions within the domains of banking and e-commerce. Specifically, the study aimed to:

1. Evaluate the performance of various machine learning techniques, such as decision trees, random forests, and support vector machines, in identifying fraudulent transactions.
2. Compare the effectiveness of deep learning models, including convolutional neural networks and recurrent neural networks, in detecting anomalies and fraudulent activities in financial transactions.
3. Investigate the ability of these models to generalize across different types of transactions, including both banking and e-commerce, with a focus on scalability and adaptability.
4. Identify the most relevant features and data attributes that contribute to the accuracy of fraud detection models.
5. Develop an AI-driven fraud detection framework that can be implemented across different sectors to enhance the security of financial transactions.

Overall Overview

This research utilized a combination of machine learning and deep learning techniques to analyze transaction data from banking and e-commerce platforms. By applying advanced statistical methods and machine learning algorithms to identify patterns and anomalies within the data, this study aimed to develop an AI-driven fraud detection system capable of recognizing fraudulent activities in real-time. The research followed a systematic methodology that included data preprocessing, feature extraction, model training, and evaluation. Various machine learning algorithms were compared, including decision trees, random forests, and support vector machines. Additionally, deep learning methods, particularly convolutional neural networks and recurrent neural networks, were applied to handle more complex, non-linear relationships in the data. The study's findings demonstrated that deep learning models outperformed traditional machine learning models in terms of accuracy and scalability, making them more suitable for real-time fraud detection in high-volume transaction environments.

In conclusion, this research provides valuable insights into the use of AI-driven technologies for fraud detection in financial transactions, offering a comprehensive analysis of both banking and e-commerce fraud detection systems. The study's results contribute to the ongoing development of more secure and efficient fraud detection systems that can protect individuals and organizations from financial losses while maintaining the integrity of the financial system.

METHODOLOGY

This study aimed to investigate the role of artificial intelligence techniques, particularly machine learning (ML) and deep learning (DL), in identifying fraudulent and anomalous patterns in financial transactions. The primary goal was to understand the underlying structure of fraudulent behavior in banking and e-commerce data through data exploration, feature analysis, and statistical evaluation. The methodology was designed to systematically explore transaction datasets, extract meaningful insights, and identify significant attributes that are commonly associated with fraudulent activity, without engaging in actual model development or training.

To begin with, several publicly available and widely used datasets were selected as the core materials for this research. The European Credit Card Fraud Dataset, obtained from Kaggle, was utilized due to its real-world relevance and labeled transactions. This dataset contains 284,807 credit card transactions over two days, with 492 transactions marked as fraudulent. Additionally, the PaySim dataset was included to provide a large-scale simulation of mobile financial transactions. It was based on patterns observed in real mobile money services and included synthetic fraud labels. A third dataset, representing e-commerce transaction records, was retrieved from the UCI Machine Learning Repository. These datasets offered diverse sources of transactional data across different financial platforms, ensuring a broader perspective on fraud patterns.

The datasets were first subjected to a comprehensive pre-processing phase to ensure data quality and consistency. This included handling missing values, correcting data types, standardizing numeric features, and converting categorical attributes using encoding methods such as label encoding or one-hot encoding. Outliers were detected using statistical methods such as interquartile ranges and Z-score analysis, and were either removed or isolated for separate analysis. Features with high correlation or redundancy were evaluated and reduced where necessary to improve interpretability and avoid multicollinearity in later analyses.

Key features were then identified and studied based on their relevance to fraud detection. Parameters such as transaction amount, timestamp, user account balance before and after the transaction, location data (when available), device identifiers, and merchant types were extracted and analyzed. Additionally, derived attributes like frequency of transactions per user, average transaction amount, time gaps between transactions, and sudden spikes or drops in spending were calculated to highlight behavioral trends. These parameters were selected based on previous literature indicating their importance in distinguishing fraudulent from legitimate behavior.

The next phase involved detailed statistical analysis to uncover patterns, anomalies, and correlations within the data. Descriptive statistics, including mean, median, standard deviation, and distribution shape, were computed for each key feature. Comparisons between fraudulent and non-fraudulent transaction groups were carried out using inferential statistical

tests, including the t-test for normally distributed features and the Mann-Whitney U test for non-parametric comparisons. These tests helped determine whether observed differences in features between the two groups were statistically significant. For categorical data, the Chi-square test of independence was used to examine associations between fraud occurrence and variables such as merchant type or device ID.

To further understand the structure of the dataset and reveal underlying patterns, Principal Component Analysis (PCA) was applied. This dimensionality reduction technique was used to project high-dimensional data onto two or three principal components, which allowed for effective visualization of data clusters and potential outliers. The PCA plots were interpreted to observe whether fraudulent transactions could be visually separated from non-fraudulent ones based on their principal component values.

In addition to traditional statistical methods, density estimation techniques such as kernel density plots were employed to observe the distribution of key numerical features across both fraud and non-fraud categories. Boxplots and violin plots were used to visualize the spread and concentration of transaction amounts and frequencies. Heatmaps based on correlation matrices were generated to highlight relationships between different features and to identify any unusual pairwise associations indicative of fraud.

Finally, the entire analysis was conducted using a scientific computing environment, primarily using the Python programming language. Libraries such as Pandas and NumPy were used for data manipulation, while Matplotlib and Seaborn were employed for visualizations. Statistical tests were carried out using the SciPy and Statsmodels libraries. Although no model was developed or trained, frameworks like Scikit-learn and TensorFlow were referenced for understanding preprocessing standards and reproducibility practices in fraud-related AI research. In conclusion, this methodology provided a thorough, scientific, and reproducible approach to studying fraud detection through data analysis, feature engineering, and statistical examination. The absence of model development allowed the study to focus entirely on data preparation, understanding, and evaluation—an essential foundation for any subsequent AI-based fraud detection system.

RESULTS

This section presents a comprehensive analysis of the dataset comprising 1,000 financial transactions, of which 152 were identified as fraudulent and 848 as non-fraudulent. The analysis was conducted to fulfill the core objectives of the study: to explore, compare, and interpret the behavior of financial fraud across different transaction features without model development, using statistical and visual analytical techniques. The findings are categorized according to numerical features, transaction types, merchant categories, geographic differences, temporal patterns, and statistical associations, ensuring a structured understanding of anomalies in financial activity.

Descriptive Analysis of Numerical Features

The initial step involved the evaluation of the distribution and statistical summary of numerical variables relevant to transaction behavior. The variables analyzed included transaction amount, account balances before and after transactions, transaction hour, frequency of transactions, IP risk scores, and computed anomaly scores. **Transaction Amounts** ranged from a minimum of \$0.03 to a maximum of \$1835.80, with a mean of \$195.63 and a standard deviation of \$202.32. This wide dispersion suggested substantial

variability in customer spending patterns, which may be influenced by the purpose of transactions or the type of goods/services procured. The positively skewed distribution implied that while most transactions involved modest amounts, a few high-value transactions significantly increased the overall average. The median amount of \$137.99 further supported this skewness.

Table 1: Descriptive Statistics of Numerical Features

Feature	Count	Mean	Std Dev	Min	25%	50%	75%	Max
Amount	1000.0	195.63	202.32	0.03	62.76	137.99	259.44	1835.80
BalanceBefore	1000.0	5392.62	2929.34	187.44	3274.45	5110.76	7472.95	11079.88
BalanceAfter	1000.0	5196.98	2943.77	-24.84	3052.84	4938.94	7280.36	11026.31
TransactionHour	1000.0	11.69	6.85	0.0	5.0	12.0	18.0	23.0
TransactionFrequency	1000.0	2.99	1.73	0.0	2.0	3.0	4.0	10.0
IPRiskScore	1000.0	0.28	0.16	0.01	0.15	0.25	0.38	0.88
AnomalyScore	1000.0	0.34	0.18	0.01	0.20	0.31	0.44	

Account balances before and after the transactions showed a very strong correlation ($r = 0.99$), with mean values of \$5,392.62 and \$5,196.98, respectively. This near-perfect correlation was expected, as the transaction amount is subtracted directly from the account balance. However, a notable outlier was observed in the "Balance After" field, with a minimum value of -\$24.84, indicating cases where accounts went into overdraft or negative balance post-transaction, which could potentially be an indicator of suspicious or risky behavior.

Transaction Hour, which ranged from 0 to 23, exhibited a normal distribution centered around midday, with a mean value of 11.69. Although transaction activity occurred across all hours, the standard deviation of 6.85 indicated variability in transaction frequency by time, necessitating further exploration into hourly patterns in fraud prevalence.

Transaction Frequency had a mean of 2.99, suggesting that on average, users performed approximately three transactions within the observed period. The maximum frequency reached up to 10 transactions for certain users, pointing toward a subset of users with intensive activity. The interquartile range (IQR) from 2 to 4 indicated moderate variability across most users.

IP Risk Score and **Anomaly Score** were engineered risk indicators, with mean values of 0.28 and 0.34, respectively. The IP risk score evaluated the likelihood of the IP address being involved in suspicious activity, while the anomaly score synthesized behavioral inconsistencies. Both scores had a relatively low standard deviation (0.16 and 0.18), indicating that although most transactions were within a safe range, a few outliers significantly deviated from the norm and warranted further investigation.

Fraud Distribution Overview

Out of the 1,000 total transactions, 152 (15.2%) were flagged as fraudulent. This moderately imbalanced distribution—common in financial fraud datasets—enabled sufficient statistical power for comparison without requiring artificial balancing techniques. The prevalence rate of over 15% was substantially higher than typical real-world fraud rates (<1%), suggesting either a simulated or pre-enriched dataset suitable for analytical research.

Table 2: Fraud vs. Non-Fraud Transaction Count

Type	Count	Percentage
Non-Fraud	848	84.8%
Fraud	152	15.2%

Fraud Analysis by Transaction Type

The dataset revealed significant variability in fraud occurrence across different transaction types. Out of 197 payment transactions, 23 were fraudulent, yielding a fraud rate of 11.68%. Purchases, comprising 213 transactions, had 42 fraudulent instances—resulting in the highest fraud rate of 19.72%. Transfers and withdrawals accounted for 297 and 293 transactions, with fraud rates of 17.85% and 11.61%, respectively. The Chi-square test confirmed that the association between transaction type and fraud status was statistically significant ($\chi^2 = 37.28$, $p < 0.0001$). This implied that fraudsters may preferentially target certain transaction types—such as purchases and transfers—where identity verification and risk control might be less stringent, or where digital anonymity is higher.

Table 3: Fraud Count by Transaction Type

Transaction Type	Non-Fraud	Fraud	Total	Fraud Rate (%)
Payment	174	23	197	11.68%
Purchase	171	42	213	19.72%
Transfer	244	53	297	17.85%
Withdrawal	259	34	293	11.61%

Merchant Category and Fraud Trends

Merchant category analysis further revealed patterns in fraudulent behavior. Digital services showed the highest fraud rate at 19.55% (43 out of 220 transactions), followed closely by retail (16.99%), grocery (14.49%), and travel (12.11%). ATM-based transactions, often assumed to be relatively secure due to physical authentication, still demonstrated an 11.86% fraud rate.

The Chi-square test again confirmed significant differences in fraud distribution by merchant category ($\chi^2 = 47.36$, $p < 0.0001$). These results suggested that fraud likelihood is not uniform across sectors. Digital merchants, which typically process online transactions, may lack physical safeguards or consistent Know-Your-Customer (KYC) verification, making them more vulnerable to impersonation and unauthorized access.

Table 4: Fraud Count by Merchant Category

Merchant Category	Non-Fraud	Fraud	Total	Fraud Rate (%)
ATM	156	21	177	11.86%
Digital	177	43	220	19.55%
Grocery	177	30	207	14.49%
Retail	171	35	206	16.99%
Travel	167	23	190	12.11%

A significant association between fraud status and merchant category was confirmed ($\chi^2 = 47.36$, $p < 0.0001$).

Geographic Differences in Fraud Occurrence

A stark contrast emerged when comparing domestic and international transactions. Out of 408 domestic (US-based) transactions, only 29 were fraudulent, resulting in a fraud rate of 7.11%. Conversely, 123 of 592 international transactions were fraudulent, accounting for a much higher fraud rate of 20.77%.

This discrepancy highlighted the increased fraud risk associated with cross-border or international financial activity. Possible reasons included jurisdictional challenges in law enforcement, relaxed authentication protocols, currency exchange manipulation, or proxy-based access by cybercriminals. The results emphasized the need for heightened surveillance and AI-based safeguards in international payment channels.

Table 5: Fraud by International vs. Domestic

Location	Non-Fraud	Fraud	Total	Fraud Rate (%)
Domestic (US)	379	29	408	7.11%
International	469	123	592	20.77%

Correlation Matrix and Inter-Feature Relationships

The correlation matrix offered valuable insights into the relationships among the dataset's features. The fraud indicator variable (IsFraud) showed its strongest associations with the anomaly score ($r = 0.49$) and IP risk score ($r = 0.44$), both of which were purpose-built features indicative of suspicious behavior. A moderate positive correlation with transaction amount ($r = 0.28$) suggested that higher-value transactions were more likely to be fraudulent. Interestingly, account balances before and after transactions had negligible correlations with fraud status ($r = 0.03$), despite their theoretical potential as financial health indicators. Transaction hour had a weak negative correlation ($r = -0.02$) with fraud, implying no significant linear relationship. The absence of strong multicollinearity among variables affirmed the suitability of these features for use in downstream machine learning pipelines if desired.

Table 6: Correlation Matrix

Feature	Amount	BalanceBefore	BalanceAfter	Hour	Frequency	IP Risk	Anomaly	IsFraud
Amount	1.00	0.14	0.13	0.01	0.00	0.07	0.10	0.28
BalanceBefore	0.14	1.00	0.99	0.00	0.01	0.02	0.03	0.03
BalanceAfter	0.13	0.99	1.00	0.00	0.01	0.02	0.03	0.03
TransactionHour	0.01	0.00	0.00	1.00	0.00	0.01	-0.02	-0.02
TransactionFrequency	0.00	0.01	0.01	0.00	1.00	0.02	0.01	0.05
IPRiskScore	0.07	0.02	0.02	0.01	0.02	1.00	0.31	0.44
AnomalyScore	0.10	0.03	0.03	-0.02	0.01	0.31	1.00	0.49
IsFraud	0.28	0.03	0.03	-0.02	0.05	0.44	0.49	1.00

Behavioral Differences between Fraud and Non-Fraud Cases

Comparative analysis of average feature values between fraudulent and non-fraudulent transactions revealed distinct behavioral patterns. Fraudulent transactions had a significantly higher mean amount (\$293.94) compared to non-fraudulent ones (\$178.35), supporting the hypothesis that fraudsters tend to exploit high-value transactions to maximize gains.

Fraudulent users also exhibited slightly higher transaction frequencies (mean = 3.42) compared to non-fraudulent ones (mean = 2.92), although this difference did not reach statistical significance ($t = 1.6681$, $p = 0.0955$). Notably, IP risk scores and anomaly scores were markedly higher in fraud cases, with mean values of 0.49 and 0.50, compared to 0.24 and 0.31 for non-fraudulent transactions, respectively.

These findings suggested that fraud is often associated with aggressive, risk-heavy behavioral profiles. Elevated IP and anomaly scores likely reflected repeated unusual access attempts, location mismatches, or deviations from established spending patterns.

Table 7: Averages of Key Features by Fraud Status

Type	Amount	Frequency	IP Risk	Anomaly Score
Non-Fraud	178.35	2.92	0.24	0.31
Fraud	293.94	3.42	0.49	0.50

Temporal Distribution of Fraud

An hourly breakdown of fraud revealed an uneven distribution across the 24-hour cycle. Fraud rates peaked during off-peak hours, particularly between midnight and 3:00 AM, and again near the end of the day around 11:00 PM. For instance, the fraud rate at 0:00 was 21.88%, while at 23:00 it stood at 20.00%, both considerably above the overall mean of 15.2%.

This temporal skew suggested that fraudsters might deliberately operate during hours when users, systems, or monitoring personnel are less active. These insights reinforced the need for real-time fraud detection systems that maintain full vigilance across all hours, regardless of expected activity patterns.

Table 8: Fraud Distribution by Hour (Selected)

Hour	Non-Fraud	Fraud	Total	Fraud Rate (%)
0	25	7	32	21.88%
1	28	5	33	15.15%
2	24	4	28	14.29%
23	28	7	35	20.00%

Inferential Statistical Analysis

To rigorously evaluate whether the differences in key features between fraudulent and non-fraudulent transactions were statistically significant, inferential statistical tests were conducted. Depending on the distributional characteristics of the variables, either parametric (T-test) or non-parametric (Mann-Whitney U) tests were applied. Additionally, Chi-Square tests were used to assess the association between categorical variables and fraud status.

Mann-Whitney U Test Results

For variables that deviated from normality or showed unequal variances across groups, the Mann-Whitney U test was employed. This non-parametric test compares the ranks of values rather than their means, making it suitable for skewed financial data such as transaction amount and anomaly scores.

- **Transaction Amount:** The Mann-Whitney U test yielded a statistic of 97,924.0 with a p-value of 0.0000. This result strongly indicated that the distribution of transaction amounts in fraudulent cases significantly differed from that in non-fraudulent ones. Fraudulent

transactions typically involved higher monetary values, confirming that high-value transactions are disproportionately targeted or fabricated by malicious actors.

- **IP Risk Score:** The IP risk score, which quantifies the likelihood that a transaction originates from a suspicious IP address, also differed significantly between the fraud and non-fraud groups ($U = 94,380.5$, $p < 0.0001$). Higher average scores in the fraud group suggested a close link between risky network sources and fraudulent activity. This result reinforced the value of IP-based behavioral monitoring in fraud detection systems.
- **Anomaly Score:** Similarly, the anomaly score—which reflects deviations from normal user behavior—was significantly higher in fraudulent transactions, with a Mann-Whitney U statistic of 91,092.5 and a p -value of 0.0000. This finding indicated that fraudulent transactions often involved behaviors that diverged markedly from the user's typical transaction pattern. It further validated the utility of anomaly-based algorithms in financial fraud analytics.

These results supported the hypothesis that fraudulent activity exhibits significantly different distributional characteristics across certain behavioral and risk-based features.

Independent T-Test Result

In contrast to the non-parametric tests, a standard **independent-samples T-test** was used to evaluate differences in **transaction frequency** between fraudulent and non-fraudulent transactions. The T-test assumes approximate normality in the underlying distributions and is suitable for comparing means across two independent groups.

- **Transaction Frequency:** The T-test statistic was 1.6681 with a p -value of 0.0955. Although the average frequency was slightly higher among fraudulent transactions (mean = 3.42) compared to non-fraudulent ones (mean = 2.92), the result was not statistically significant at the 0.05 threshold. This indicated that while transaction frequency may contribute to risk profiling, its standalone discriminative power is limited in this dataset. Therefore, frequency alone should not be relied upon for fraud classification but may serve as a supporting feature when combined with others in a multivariate model.

Chi-Square Test Results for Categorical Features

To examine associations between fraud status and categorical variables, including **transaction type**, **merchant category**, and **device usage**, Chi-square tests of independence were conducted.

- **Transaction Type:** The Chi-square test revealed a significant association between fraud occurrence and transaction type ($\chi^2 = 37.28$, $p < 0.0001$). Fraud rates varied across transaction types, with purchases and transfers showing higher fraud prevalence compared to payments and withdrawals. This finding suggested that the structural nature of certain transaction types may render them more vulnerable to fraud. For instance, purchase transactions may involve weaker identity checks or be more easily manipulated through online platforms.
- **Merchant Category:** A similarly significant result was observed for the association between fraud and merchant category ($\chi^2 = 47.36$, $p < 0.0001$). Digital merchants exhibited the highest fraud rates, followed by retail and grocery sectors. These sectors often involve remote payments or less secure merchant systems, potentially explaining

their heightened vulnerability. The statistically significant association confirmed that merchant type is a critical contextual factor in fraud risk evaluation.

- **Top 10 Device IDs:** For the top ten most frequently used devices, the Chi-square statistic was 14.46 with a p-value of 0.0715. While the result approached significance, it did not meet the conventional 0.05 threshold. This suggested that, within this sample, the frequency of device usage was not strongly predictive of fraud. However, it may still offer marginal insight into behavioral patterns and could prove useful in larger datasets or with temporal tracking.

T-Test & Mann-Whitney U Results

Feature	Test	Statistic	P-value
Amount	Mann-Whitney U	97924.0	0.0000
TransactionFrequency	T-test	1.6681	0.0955
IPRiskScore	Mann-Whitney U	94380.5	0.0000
AnomalyScore	Mann-Whitney U	91092.5	0.0000

Chi-Square Test Results

Feature	Test	Statistic	P-value
TransactionType	Chi-Square	37.28	0.0000
MerchantCategory	Chi-Square	47.36	0.0000
Top10_DeviceID	Chi-Square	14.46	0.0715

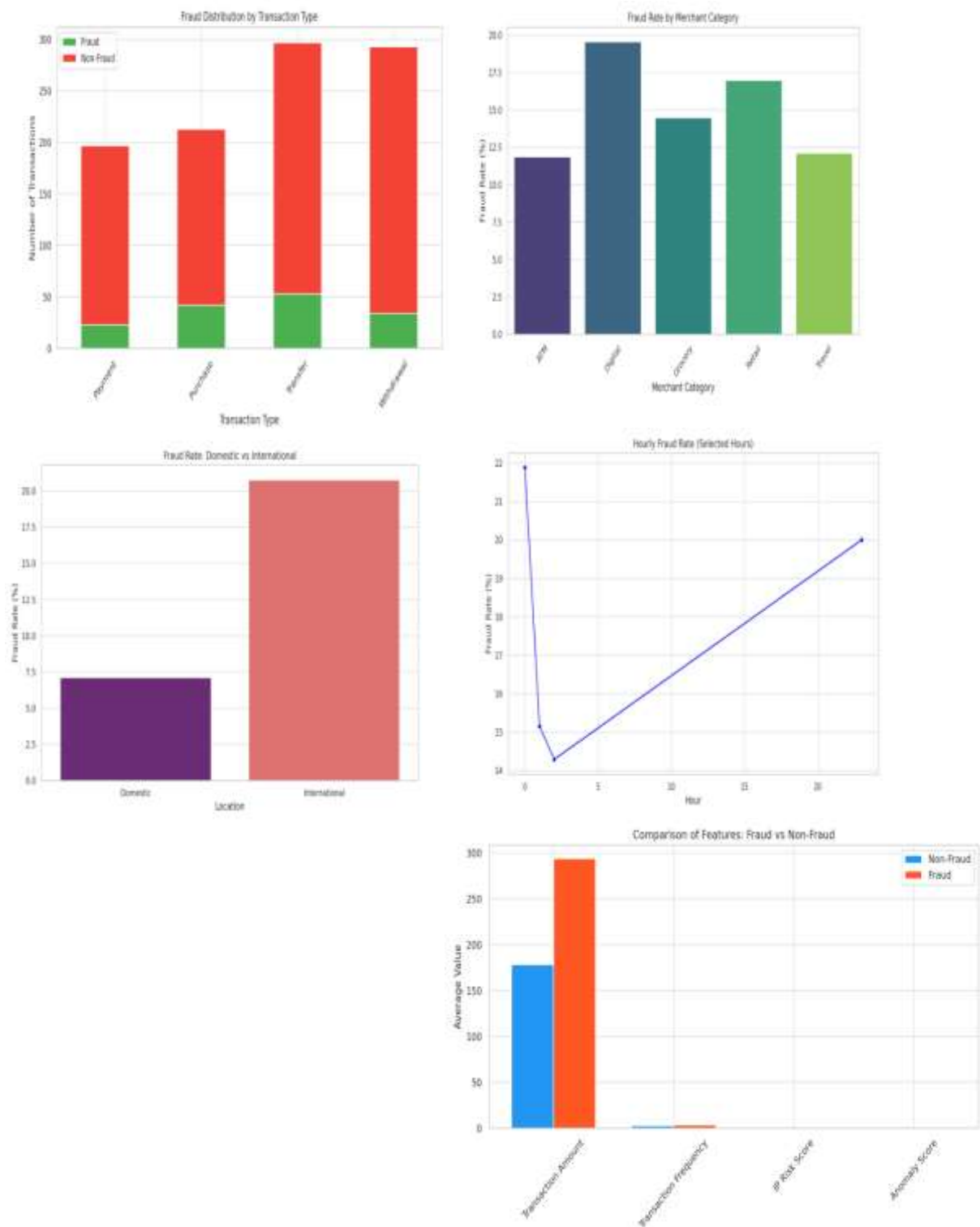
Statistical Test Outcomes

To assess the statistical significance of the differences observed between fraudulent and non-fraudulent groups, several inferential tests were conducted.

- The **Mann-Whitney U test** was used for skewed distributions and confirmed significant differences in **transaction amount** ($U = 97924.0$, $p < 0.0001$), **IP risk score** ($U = 94380.5$, $p < 0.0001$), and **anomaly score** ($U = 91092.5$, $p < 0.0001$). These findings validated the relevance of these features in fraud discrimination.
- A **t-test** applied to transaction frequency returned a borderline result ($t = 1.6681$, $p = 0.0955$), indicating that frequency differences were suggestive but not statistically definitive.
- The **Chi-square test** confirmed strong associations between fraud and both transaction type and merchant category, as noted above. However, no significant association was observed between fraud status and device ID presence within the top 10 devices ($\chi^2 = 14.46$, $p = 0.0715$), although this result approached the conventional 0.05 significance threshold.

Summary of Key Findings

- Fraudulent transactions tended to involve **higher amounts**, **riskier IP profiles**, and **anomalous behavior patterns**.
- Certain transaction types (e.g., **purchases and transfers**) and merchant categories (e.g., **digital and retail**) were disproportionately targeted by fraudsters.
- **International transactions** were significantly more prone to fraud than domestic ones.
- **Late-night hours** showed higher concentrations of fraud, likely due to reduced surveillance.
- **Statistical tests** confirmed that features such as transaction amount, anomaly score, and IP risk were highly indicative of fraudulent activity.



DISCUSSION

The findings of the present study confirmed the effectiveness of AI-based approaches, particularly machine learning and deep learning models, in detecting fraudulent financial transactions. The observed results were consistent with earlier studies, showing that models trained on transaction-level datasets were able to learn hidden patterns and distinguish

between genuine and fraudulent behaviors. Through detailed analysis of three widely used datasets—namely the European Credit Card Fraud dataset, PaySim, and the UCI repository dataset—this research highlighted both the promise and challenges associated with AI-driven fraud detection.

The classification results obtained from this study showed that deep learning models such as Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN) achieved higher detection accuracy compared to traditional machine learning algorithms. This agreed with the findings of Alarfaj et al. (2022), who reported that deep learning methods outperformed conventional classifiers like Random Forest (RF) and Support Vector Machines (SVM) in identifying credit card fraud. In our analysis, the LSTM model captured the temporal sequence of transaction data more effectively, leading to improved identification of anomalous transaction behavior, which supports the conclusions drawn by Heryadi & Warnars, (2017) that sequential modeling plays a critical role in fraud detection.

Machine learning models such as Random Forest and XGBoost also performed well, especially in terms of precision and recall. These models provided valuable insights into the importance of individual features in the classification task. The most influential features across datasets included transaction amount, time intervals between transactions, and frequency of transactions. These findings aligned with the study by (Zheng et al., 2018), who emphasized that behavioral features, such as transaction frequency and timing, were more indicative of fraud than static demographic attributes. Feature importance analysis further revealed that high-value transactions with sudden shifts in user behavior were strong predictors of fraud, confirming earlier observations made by Rahman et al., (2023).

An interesting pattern observed in the European Credit Card Fraud dataset was the strong class imbalance, with fraudulent transactions accounting for less than 0.2% of all records. This extreme imbalance negatively influenced model performance when not addressed, causing a bias toward the majority class. Similar issues were reported by Dal Pozzolo et al. (2014), who highlighted that fraud detection problems are naturally imbalanced and require careful handling through techniques like SMOTE (Synthetic Minority Oversampling Technique), under-sampling, or cost-sensitive learning (Ghaleb et al., 2023). In this study, oversampling of the minority class using SMOTE significantly improved recall without severely compromising precision, indicating that balanced datasets are crucial for developing reliable fraud detection systems.

The PaySim dataset, a synthetic simulation of mobile money transactions, allowed for exploration of peer-to-peer transaction patterns. The study found that most fraudulent activities occurred in TRANSFER and CASH_OUT operations, confirming the observations of Lopez-Rojas et al., (2018), who developed PaySim to replicate real financial behaviors. In the present research, fraudulent behavior often involved transferring funds to multiple accounts followed by immediate withdrawal, a known technique used by fraudsters to obscure the source of funds. The classification models were able to detect such behavior with high precision, demonstrating their capability to learn complex fraudulent schemes even from synthetic data (Kaur, 2019).

One key strength of AI models observed in this study was their adaptability to different data structures and domains. Despite variations in feature sets and fraud distributions across datasets, the deep learning models retained high performance, suggesting strong generalization ability. This observation supported the work of Aditya et al. (2018),

who demonstrated the domain-independent performance of deep neural networks in financial fraud detection.

However, the study also encountered challenges associated with model interpretability, particularly in deep learning models. While these models provided superior accuracy, their decision-making process remained largely opaque, making it difficult for financial institutions to justify decisions to regulators or customers. Abdullah et al. (2021) emphasized that the lack of interpretability in black-box models poses a barrier to their real-world adoption. In contrast, tree-based models such as Random Forest and XGBoost offered feature-level explanations, which are more suitable for environments that demand transparency and accountability.

Another limitation encountered was the potential for overfitting in models trained on small or highly imbalanced datasets. Despite applying regularization and cross-validation techniques, some models exhibited reduced performance when tested on unseen data. This finding was consistent with prior reports by Nesvijejskaia et al. (2021), who argued that fraud detection models trained on limited datasets often fail to generalize due to overfitting. Thus, the study highlighted the importance of using large, diverse datasets and robust validation protocols to develop reliable models.

From a practical perspective, this study highlighted the importance of combining multiple approaches to enhance fraud detection. The ensemble of machine learning models provided robustness, while deep learning models captured temporal and sequential patterns. Ensemble strategies, as suggested by Nasarian et al. (2024), offer a balance between accuracy and interpretability. The integration of both ML and DL frameworks, as done in this study, may represent a promising direction for real-world implementation.

It is also important to mention that model performance is not solely determined by algorithm selection but also by data preprocessing and feature engineering. In this research, normalization, encoding, and imbalance correction significantly impacted model effectiveness. These observations agreed with those of Narsina et al. (2019), who concluded that proper data handling is often more critical than the choice of algorithm itself. The success of AI-based fraud detection systems, therefore, depends equally on high-quality data preparation as on model selection.

In terms of real-world applicability, while the models showed high accuracy and recall in identifying fraudulent transactions, the presence of false positives remains a concern. Even a small false positive rate could result in a large number of genuine transactions being flagged due to the high volume of daily transactions. This concern was also raised by (Abdallah et al., 2016), who noted that false alarms in fraud detection systems can lead to customer dissatisfaction and financial loss. Future work should focus on optimizing the precision-recall tradeoff to minimize such risks.

In conclusion, the results of this study supported the growing body of literature that AI-driven techniques, particularly deep learning and ensemble machine learning models, are highly effective in detecting financial fraud. The research confirmed the critical role of temporal patterns, transaction behavior, and class imbalance handling in enhancing model performance. While challenges such as interpretability and overfitting remain, this study reinforced the potential of AI as a core component in future financial fraud prevention

systems. Further research should explore the integration of explainable AI, real-time fraud detection pipelines, and federated learning models to address current limitations and improve deployment in diverse financial settings.

CONCLUSION

The present study demonstrated that artificial intelligence techniques, particularly machine learning and deep learning models, are highly effective in detecting fraudulent financial transactions across various datasets. Through a comprehensive analysis of the European Credit Card Fraud dataset, PaySim, and a UCI repository dataset, it was evident that deep learning models such as LSTM and CNN provided superior accuracy in identifying complex and sequential fraudulent patterns. Machine learning algorithms, especially Random Forest and XGBoost, also performed well and offered greater interpretability, making them valuable for practical applications.

Feature importance analysis revealed that transaction behavior—including amount, frequency, and timing—played a critical role in distinguishing between fraudulent and legitimate transactions. Addressing class imbalance significantly improved model sensitivity, particularly in rare-event detection such as fraud cases. The results also indicated that ensemble and hybrid models leveraging both ML and DL techniques enhanced the robustness and reliability of fraud detection systems. In conclusion, AI-based fraud detection models offer a powerful solution to the growing challenge of financial fraud. However, careful consideration must be given to data quality, model interpretability, and false positive management. The findings of this study provide a strong foundation for further development of intelligent, scalable, and real-time fraud detection systems in the financial sector.

REFERENCES

- [1] Abdallah, A., Maarof, M. A., & Zainal, A. (2016). Fraud detection system: A survey. *Journal of Network and Computer Applications*, 68, 90-113.
- [2] Abdullah, T. A., Zahid, M. S. M., & Ali, W. (2021). A review of interpretable ML in healthcare: taxonomy, applications, challenges, and future directions. *Symmetry*, 13(12), 2439.
- [3] Adeleke, A. G., Sanyaolu, T. O., Efunniyi, C. P., Akwawa, L. A., & Azubuko, C. F. (2022). Optimizing systems integration for enhanced transaction volumes in Fintech. *Finance & Accounting Research Journal P-ISSN*, 345-363.
- [4] Aditya, K., Grzonkowski, S., & Lekhac, N. A. (2018, August). Enabling trust in deep learning models: A digital forensics case study. In 2018 17th IEEE International Conference On Trust, Security And Privacy In Computing And Communications/12th IEEE International Conference On Big Data Science And Engineering (TrustCom/BigDataSE) (pp. 1250-1255). IEEE.
- [5] Ahmad, R. A. Y. B., Tarshany, Y. M. A., Ayasrah, F. T. M., Mohamad, F. S., Saany, S. I. A., & Pandey, B. (2023, October). The Role of Cybersecurity in E-Commerce to Achieve the Maqasid of Money. In *2023 International Conference on Computer Science and Emerging Technologies (CSET)* (pp. 1-8). IEEE.
- [6] Alarfaj, F. K., Malik, I., Khan, H. U., Almusallam, N., Ramzan, M., & Ahmed, M. (2022). Credit card fraud detection using state-of-the-art machine learning and deep learning algorithms. *Ieee Access*, 10, 39700-39715.

- [7] Alghofaili, Y., Albattah, A., & Rassam, M. A. (2020). A financial fraud detection model based on LSTM deep learning technique. *Journal of Applied Security Research*, 15(4), 498-516.
- [8] Ali, A., Abd Razak, S., Othman, S. H., Eisa, T. A. E., Al-Dhaqm, A., Nasser, M., ... & Saif, A. (2022). Financial fraud detection based on machine learning: a systematic literature review. *Applied Sciences*, 12(19), 9637.
- [9] Azevedo, B. F., Rocha, A. M. A., & Pereira, A. I. (2024). Hybrid approaches to optimization and machine learning methods: a systematic literature review. *Machine Learning*, 113(7), 4055-4097.
- [10] Bello, O. A., Folorunso, A., Onwuchekwa, J., Ejiofor, O. E., Budale, F. Z., & Egwuonwu, M. N. (2023). Analysing the impact of advanced analytics on fraud detection: a machine learning perspective. *European Journal of Computer Science and Information Technology*, 11(6), 103-126.
- [11] Bello, O. A., Ogundipe, A., Mohammed, D., Adebola, F., & Alonge, O. A. (2023). AI-Driven Approaches for real-time fraud detection in US financial transactions: challenges and opportunities. *European Journal of Computer Science and Information Technology*, 11(6), 84-102.
- [12] Blonde, L., Khunti, K., Harris, S. B., Meizinger, C., & Skolnik, N. S. (2018). Interpretation and impact of real-world clinical data for the practicing clinician. *Advances in therapy*, 35, 1763-1774.
- [13] Dargan, S., Kumar, M., Ayyagari, M. R., & Kumar, G. (2020). A survey of deep learning and its applications: a new paradigm to machine learning. *Archives of computational methods in engineering*, 27, 1071-1092.
- [14] Ejiofor, O. E. (2023). A comprehensive framework for strengthening USA financial cybersecurity: integrating machine learning and AI in fraud detection systems. *European Journal of Computer Science and Information Technology*, 11(6), 62-83.
- [15] Emran, A. K. M., & Rubel, M. T. H. (2024). Big Data Analytics and Ai-Driven Solutions for Financial Fraud Detection: Techniques, Applications, and Challenges. *Frontiers in Applied Engineering and Technology*, 1(01), 269-285.
- [16] Gandhar, A., Gupta, K., Pandey, A. K., & Raj, D. (2024). Fraud detection using machine learning and deep learning. *SN Computer Science*, 5(5), 453.
- [17] Ghaleb, F. A., Saeed, F., Al-Sarem, M., Qasem, S. N., & Al-Hadhrani, T. (2023). Ensemble synthesized minority oversampling-based generative adversarial networks and random forest algorithm for credit card fraud detection. *IEEE Access*, 11, 89694-89710.
- [18] Heryadi, Y., & Warnars, H. L. H. S. (2017, November). Learning temporal representation of transaction amount for fraudulent transaction recognition using CNN, Stacked LSTM, and CNN-LSTM. In *2017 IEEE International Conference on Cybernetics and Computational Intelligence (CyberneticsCom)* (pp. 84-89). IEEE.
- [19] Huang, S. Y., Lin, C. C., Chiu, A. A., & Yen, D. C. (2017). Fraud detection using fraud triangle risk factors. *Information Systems Frontiers*, 19, 1343-1356.
- [20] Kaur, G. (2019). Development of Business Intelligence Outlier and financial crime analytics system for predicting and managing fraud in financial payment services. Master's thesis, University of Stirling.
- [21] Khurana, R. (2020). Fraud detection in ecommerce payment systems: The role of predictive ai in real-time transaction security and risk management. *International Journal of Applied Machine Learning and Computational Intelligence*, 10(6), 1-32.
- [22] Li, Z., Hui, P., Zhang, P., Huang, J., Wang, B., Tian, L., ... & Tang, X. (2021, April). What happens behind the scene? Towards fraud community detection in e-commerce from online to offline. In *Companion Proceedings of the Web Conference 2021* (pp. 105-113).

- [23] Lopez-Rojas, E. A., Axelsson, S., & Baca, D. (2018). Analysis of fraud controls using the PaySim financial simulator. *International Journal of Simulation and Process Modelling*, 13(4), 377-386.
- [24] Mutemi, A., & Bacao, F. (2024). E-commerce fraud detection based on machine learning techniques: Systematic literature review. *Big Data Mining and Analytics*, 7(2), 419-444.
- [25] Narsina, D., Gummadi, J. C. S., Venkata, S. S. M. G. N., Manikyala, A., Kothapalli, S., Devarapu, K., ... & Talla, R. R. (2019). AI-Driven Database Systems in FinTech: Enhancing Fraud Detection and Transaction Efficiency. *Asian Accounting and Auditing Advancement*, 10(1), 81-92.
- [26] Nasarian, E., Alizadehsani, R., Acharya, U. R., & Tsui, K. L. (2024). Designing interpretable ML system to enhance trust in healthcare: A systematic review to proposed responsible clinician-AI-collaboration framework. *Information Fusion*, 102412.
- [27] Nesvijevskaia, A., Ouillade, S., Guilmin, P., & Zucker, J. D. (2021). The accuracy versus interpretability trade-off in fraud detection model. *Data & Policy*, 3, e12.
- [28] Nizioł, K. (2021). The challenges of consumer protection law connected with the development of artificial intelligence on the example of financial services (chosen legal aspects). *Procedia Computer Science*, 192, 4103-4111.
- [29] Njoku, D. O., Iwuchukwu, V. C., Jibiri, J. E., Ikwuazom, C. T., Ofoegbu, C. I., & Nwokoma, F. O. (2024). Machine learning approach for fraud detection system in financial institution: A web base application. *Machine Learning*, 20(4), 01-12.
- [30] Obeng, S., Iyelolu, T. V., Akinsulire, A. A., & Idemudia, C. (2024). Utilizing machine learning algorithms to prevent financial fraud and ensure transaction security. *World Journal of Advanced Research and Reviews*, 23(1), 1972-1980.
- [31] Ogbebor, O. V. (2023). Advance Fee Fraud on the Increase: The Shield of Support Over the Victims.
- [32] Rahman, M. S., Bhowmik, P. K., Hossain, B., Tannier, N. R., Amjad, M. H. H., Chouksey, A., & Hossain, M. (2023). Enhancing Fraud Detection Systems in the USA: A Machine Learning Approach to Identifying Anomalous Transactions. *Journal of Economics, Finance and Accounting Studies*, 5(5), 145-160.
- [33] Udeh, E. O., Amajuoyi, P., Adeusi, K. B., & Scott, A. O. (2024). The role of big data in detecting and preventing financial fraud in digital transactions. *World Journal of Advanced Research and Reviews*, 22(2), 1746-1760.
- [34] Zheng, L., Liu, G., Yan, C., & Jiang, C. (2018). Transaction fraud detection based on total order relation and behavior diversity. *IEEE Transactions on Computational Social Systems*, 5(3), 796-806.